

Irreversible Port-Hamiltonian Formulations for 1-Dimensional Fluid Systems

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Motivation

- Many physical systems involve:
 - thermodynamic irreversibility,
 - temperature-dependent dynamics,
 - transport and convection.
- Classical port-Hamiltonian systems:
 - handle energy conservation,
 - but not irreversible entropy production.
- **Irreversible Port-Hamiltonian Systems (IPHS)** provide a thermodynamically consistent framework.

Example: Gas-piston system

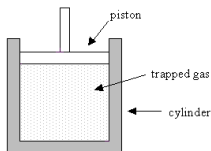


Figure: Ideal gas-piston system

The total energy of the system can be written as

$$H(x, s) := U(s, V) + H_0(p, q),$$

where $x = [q \quad p \quad V]^\top$, $H_0 := \frac{1}{2} \frac{p^2}{m} + mgq$.

The co-state (co-energy) variables are as follows

$$\frac{\partial H}{\partial q} = mg, \quad \frac{\partial H}{\partial p} = v, \quad \frac{\partial H}{\partial s} = T, \quad \frac{\partial H}{\partial V} = -P.$$

The dynamics is represented by

$$\frac{dq}{dt} = v, \quad \frac{dp}{dt} = -mg + AP - \mu v, \quad \frac{dV}{dt} = Av, \\ \frac{ds}{st} = \frac{1}{T} \mu v^2 := \sigma_{int}.$$

The state-space representation is

$$\frac{d}{dt} \begin{bmatrix} q \\ p \\ V \\ s \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & -A & -\frac{\mu v}{T} \\ 0 & A & 0 & 0 \\ 0 & \frac{\mu v}{T} & 0 & 0 \end{bmatrix} \begin{bmatrix} F_g \\ v \\ -P \\ T \end{bmatrix}.$$

IPHS Definition

Definition

An infinite-dimensional irreversible port-Hamiltonian system undergoing m irreversible processes is defined by

$$\begin{bmatrix} \partial_t x \\ \partial_t s \end{bmatrix} = \begin{bmatrix} P_0 & G_0 \mathbf{R}_0 \\ -\mathbf{R}_0^\top G_0^\top & 0 \end{bmatrix} \begin{bmatrix} e_x \\ e_s \end{bmatrix} + \begin{bmatrix} P_1 \partial_\zeta(\cdot) & \partial_\zeta(G_1 \mathbf{R}_1 \cdot) \\ -\mathbf{R}_1^\top G_1^\top \partial_\zeta(\cdot) & r_s \partial_\zeta(\cdot) + \partial_\zeta(r_s \cdot) \end{bmatrix} \begin{bmatrix} e_x \\ e_s \end{bmatrix},$$

where $e_x(\zeta, t) = \delta_x \mathcal{H}$, $e_s(\zeta, t) = \delta_s \mathcal{H}$ are the co-state variables.

$\mathbf{R}_0(\mathbf{x}), \mathbf{R}_1(\mathbf{x}) \in \mathbb{R}^{m \times 1}$, $r_s(\mathbf{x}) \in \mathbb{R}$ are the modulating functions defined as follows

$$R_{0,i} = \gamma_{0,i}(\mathbf{x}, e_x, e_s) \{S|G_0(:, i)|\mathcal{H}\},$$

$$R_{1,i} = \gamma_{1,i}(\mathbf{x}, e_x, e_s) \{S|G_1(:, i)|\partial_\zeta \mathcal{H}\},$$

$$r_s = \gamma_s(\mathbf{x}, e_x, e_s) \{S|\mathcal{H}\},$$

where $\mathbf{x} = [x^\top s]^\top$.

The pseudo Poisson brackets are defined as

$$\{\mathcal{E}|\mathcal{G}|\mathcal{F}\} = \begin{bmatrix} \delta_{\mathbf{x}} \mathcal{E} \\ \delta_s \mathcal{E} \end{bmatrix}^\top \begin{bmatrix} 0 & \mathcal{G} \\ -\mathcal{G}^* & 0 \end{bmatrix} \begin{bmatrix} \delta_{\mathbf{x}} \mathcal{F} \\ \delta_s \mathcal{F} \end{bmatrix},$$

$$\{\mathcal{E}|\mathcal{F}\} = \delta_s \mathcal{E} \partial_\zeta (\delta_s \mathcal{F})$$

for some smooth function $\mathcal{E}$ and $\mathcal{F}$, operator $\mathcal{G}$ and its formal adjoint $\mathcal{G}^*$.

Boundary inputs and outputs

The boundary inputs and outputs are defined by

$$\mathbf{u}(t) = W_B \begin{bmatrix} \tilde{\mathbf{e}}|_L \\ \tilde{\mathbf{e}}|_0 \end{bmatrix}, \quad \mathbf{y}(t) = W_C \begin{bmatrix} \tilde{\mathbf{e}}|_L \\ \tilde{\mathbf{e}}|_0 \end{bmatrix}, \quad (1)$$

where the boundary port variables are $\tilde{\mathbf{e}}(\zeta, t) = \begin{bmatrix} e_x \\ \mathbf{R}e_s \end{bmatrix}$ with $\mathbf{R} = [1 \quad \mathbf{R}_1^\top \quad r_s]^\top \in \mathbb{R}^{m+2}$.

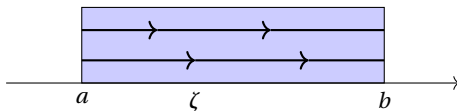
The Hamiltonian is defined as follows

$$\mathcal{H} = \int_a^b \frac{1}{2} \rho v^2 + u(\rho, s)$$

With this, the total energy balance $\dot{\mathcal{H}} = \mathbf{y}^\top \mathbf{u}$ and total entropy balance $\dot{S} = \int_a^b \sigma_s d\zeta + \mathbf{u}_s^\top \mathbf{y}_s$ are satisfied, where $\mathbf{u}_s$ and $\mathbf{y}_s$ are entropy conjugated input and output respectively. Moreover, $\sigma_s \geq 0$ represents the total internal entropy production.

Our system

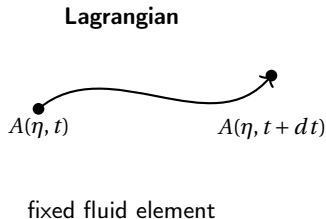
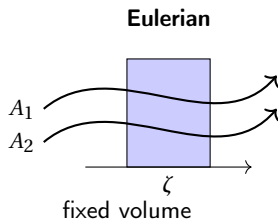
Let's consider a non-isentropic compressible fluid within a one-dimensional spatial domain, $\zeta \in [a, b]$:



We denote by

- $\rho = \rho(\zeta, t)$, the fluid mass density per unit length.
- $v = v(\zeta, t)$ velocity
- $p = p(\zeta, t)$ the pressure

Eulerian and Lagrangian reference frames



- **Eulerian frame:**
 - fixed spatial domain,
 - convective terms appear explicitly.
 - **Lagrangian frame:**
 - follows material particles,
 - convection absorbed into coordinates.
- ⇒ Both descriptions are physically equivalent but lead to different mathematical structures.

Governing Equations (Eulerian)

- Continuity equation:

$$\frac{\partial \rho}{\partial t} + \partial_{\zeta}(\rho v) = 0. \quad (2)$$

- Momentum equation:

$$\partial_t(\rho v) + \partial_{\zeta}(\rho v^2) = \partial_{\zeta} \sigma = \partial_{\zeta}(-p - \tau), \tau = \mu \partial_{\zeta} v \quad (3)$$

where p represents thermodynamic pressure, τ represents the viscous stress

By continuity equation, we have $\rho \partial_t v = -\rho v \partial_{\zeta} v - \partial_{\zeta} p + \partial_{\zeta}(\mu \partial_{\zeta} v)$. Using Gibbs' equation $du = -pd\left(\frac{1}{\rho}\right) + Tds$, $u := h + \frac{p}{\rho}$, we have

$$\frac{1}{\rho} \partial_{\zeta} p = \partial_{\zeta} h - T \partial_{\zeta} s.$$

By combining all,

$$\partial_t v = -v \partial_{\zeta} v - \partial_{\zeta} h + T \partial_{\zeta} s + \frac{1}{\rho} \partial_{\zeta}(\mu \partial_{\zeta} v) \quad (4)$$

Governing Equations (Eulerian): Continued

- Entropy equation: We can write down the rate of internal energy as

$$\partial_t u = -v \partial_\zeta u - \frac{1}{\rho} \partial_\zeta q - \frac{p}{\rho} \partial_\zeta v - \frac{\tau}{\rho} \partial_\zeta v.$$

Using Fourier's law, $q = -k \partial_\zeta T$, where k is the thermal conductivity, we have

$$\partial_t u = -v \partial_\zeta u + \frac{1}{\rho} \partial_\zeta (k \partial_\zeta T) - \frac{p}{\rho} \partial_\zeta v + \frac{\mu}{\rho} (\partial_\zeta v)^2.$$

Again, from Gibbs' equation, we have

$$D_t s = \frac{1}{T} D_t u + \frac{p}{T} D_t (1/\rho),$$

where, $D_t := \partial_t + v \partial_\zeta$ is the material derivative. Then, we have

$$\partial_t s = -v \partial_\zeta s + \frac{1}{\rho T} \partial_\zeta (k \partial_\zeta T) + \frac{\mu}{\rho T} (\partial_\zeta v)^2 \quad (5)$$

Port Hamiltonian modeling

- Energy variables: $(\rho(\zeta, t), v(\zeta, t), s(\zeta, t))$

- Hamiltonian: $\mathcal{H}(\rho, v, s) = \int_a^b \left(\frac{1}{2} \rho v^2 + \rho u \right) dx = \int_a^b \left(\frac{\rho v^2}{2} + \rho h - p \right) dx$

- Effort variables:

$$\delta_\rho \mathcal{H} = \frac{1}{2} v^2 + h, \quad \delta_v \mathcal{H} = \rho v, \quad \delta_s \mathcal{H} = \frac{\partial \mathcal{H}}{\partial u} \cdot \frac{\partial u}{\partial s} = \rho T$$

- PH formalism:

$$\begin{pmatrix} \partial_t \rho \\ \partial_t v \\ \partial_t s \end{pmatrix} = \begin{pmatrix} 0 & -\partial_\zeta & 0 \\ -\partial_\zeta & 0 & A \\ 0 & B & \frac{1}{\rho T} \partial_\zeta (k \partial_\zeta T) \end{pmatrix} \begin{pmatrix} \delta_\rho \mathcal{H} \\ \delta_v \mathcal{H} \\ \delta_s \mathcal{H} \end{pmatrix}$$

Where the operators A and B are defined from Hilbert space $H^1((a, b))$ to $H^1((a, b))$ as follows:

$$A: f \mapsto \frac{1}{\rho} \partial_\zeta s \cdot f + \frac{\mu}{\rho} \cdot \partial_\zeta \left(\frac{\partial_\zeta v}{T} \cdot \frac{f}{\rho} \right)$$

$$B: g \mapsto -\frac{1}{\rho} \partial_\zeta s \cdot g + \frac{\mu}{\rho} \cdot \frac{\partial_\zeta v}{T} \partial_\zeta \left(\frac{g}{\rho} \right)$$

B is the formal skew-adjoint operator of A .

Energy balance

$$\begin{aligned}\frac{d\mathcal{H}}{dt} &= \int_a^b \frac{\delta\mathcal{H}}{\delta\mathbf{x}} \frac{\partial\mathbf{x}}{\partial t} d\zeta \\ &= \left[-\left(\frac{1}{2}v^2 + h\right)\rho v + \mu v \partial_\zeta v + k \partial_\zeta T \right]_a^b\end{aligned}\quad (6)$$

Entropy balance: Second law of thermodynamics

$S := \int_a^b \rho s d\zeta$ is the total entropy of the system.

$$\begin{aligned}\frac{dS}{dt} &= \int_a^b (\delta_\rho S \quad \delta_v S \quad \delta_s S) \partial_t x d\zeta \\ &= - \left[s \rho v - \frac{k \partial_\zeta T}{T} \right]_a^b + \underbrace{\int_a^b \frac{1}{T^2} k (\partial_\zeta T)^2 + \frac{\mu}{T} (\partial_\zeta v)^2 d\zeta}_{\text{Internal entropy production}}\end{aligned}\quad (7)$$

From Eulerian to Lagrangian

Consider now a small fluid element as in Figure 2.

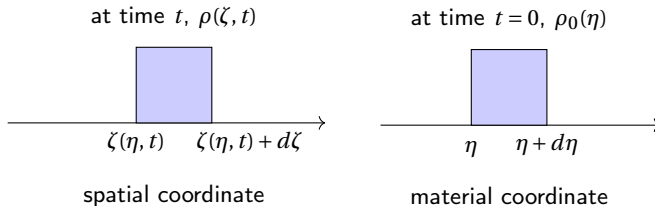


Figure: Small fluid element in spatial and material coordinates.

Mass conservation principle

$$dm_0 = dm_t \implies \rho_0(\eta) d\eta = \rho(\zeta, t) d\zeta.$$

$$\implies \frac{\partial \eta}{\partial \zeta} = \frac{\rho(\zeta, t)}{\rho_0(\eta)}.$$

Mass balance equation

$$\partial_t \rho = -\partial_\zeta(\rho v) \implies D_t \rho = \partial_t \rho + v \partial_\zeta \rho$$

$$\implies D_t \left(\frac{1}{\rho} \right) = \frac{1}{\rho_0} \partial_\eta v.$$

Governing Equations (Lagrangian)

State variables:

$$(\phi(\eta, t) = \frac{1}{\rho}(\eta, t), v(\eta, t), s(\eta, t))$$

- Continuity equation:

$$D_t \left(\frac{1}{\rho} \right) = \frac{1}{\rho_0} \partial_\eta v. \quad (8)$$

- Momentum equation:

$$D_t v = -\frac{1}{\rho_0} \partial_\eta p + \frac{\mu}{\rho_0^2} \partial_\eta^2 v. \quad (9)$$

- Entropy equation:

$$D_t s = \frac{1}{\rho_0 T} \left(k \partial_\eta^2 T + \mu (\partial_\eta v)^2 \right). \quad (10)$$

Port Hamiltonian modeling

- Hamiltonian: $H(\phi, v, s) = \int_a^b \left(\frac{1}{2} \rho v^2 + \rho u \right) d\zeta = \int_a^b \left(\frac{1}{2} v^2 + u \right) \rho_0 d\eta.$

- Effort variables:

$$\delta_\phi H = -\rho_0 p, \quad \delta_v H = \rho_0 v, \quad \delta_s H = \rho_0 T.$$

- PH formalism:

$$\begin{pmatrix} \partial_t \phi \\ \partial_t v \\ \partial_t s \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{\rho_0} \partial_\eta \left(\frac{1}{\rho_0} \cdot \right) & 0 \\ \frac{1}{\rho_0} \partial_\eta \left(\frac{1}{\rho_0} \cdot \right) & 0 & A_l \\ 0 & B_l & \frac{1}{\rho_0 T} \partial_\eta \left(\frac{k}{\phi \rho_0} \partial_\eta \left(\frac{1}{\rho_0} \delta_s H \right) \right) \end{pmatrix} \begin{pmatrix} \delta_\phi H \\ \delta_v H \\ \delta_s H \end{pmatrix} \quad (11)$$

Where the operators A_l and B_l are defined from $H^2([a, b])$ to $H^2([a, b])$ as follows:

$$A_l : f \mapsto \frac{1}{\rho_0} \partial_\eta \left(\frac{\mu}{\Phi T \rho_0^2} \partial_\eta v f \right)$$

$$B_l : g \mapsto \frac{\mu \partial_\eta v}{T \phi \rho_0^2} \partial_\eta \left(\frac{1}{\rho_0} g \right),$$

B_l is the formal skew-adjoint operator of A_l .

Energy balance: the first law of Thermodynamics

$$\begin{aligned}\frac{dH}{dt} &= \int_a^b \frac{\delta H}{\delta \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t} d\eta \\ &= [-pv + \mu v \partial_\eta v + f \partial_\eta T]_a^b\end{aligned}\quad (12)$$

Entropy balance: Second law of thermodynamics

$S := \int_a^b \rho_0 s d\eta$ is the total entropy of the system.

$$\begin{aligned}\frac{dS}{dt} &= \int_a^b \frac{\delta S}{\delta \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t} dz \\ &= \underbrace{\frac{\mu}{\Phi \rho_0 T} (\partial_\eta v)^2 + \frac{k}{\Phi \rho_0 T^2} (\partial_\eta T)^2}_{\text{Internal entropy production}} + \left[\frac{k}{T} \partial_\eta T \right]_a^b\end{aligned}\quad (13)$$

Infinite-dimensional IPHS for convective flows

A general structure of an infinite-dimensional irreversible port-Hamiltonian system with potential convective flows undergoing m irreversible processes is defined by

$$\begin{bmatrix} \partial_t x \\ \partial_t s \end{bmatrix} = \begin{bmatrix} P_0 & G_1 C(\mathbf{x}) + G_0 \mathbf{R}_0 \\ -G_1^\top C(\mathbf{x}) - \mathbf{R}_0^\top G_0^\top & 0 \end{bmatrix} \begin{bmatrix} \delta_x \mathcal{H} \\ \delta_s \mathcal{H} \end{bmatrix} + \begin{bmatrix} P_1 \partial_\zeta^{f_0}(\cdot) & G_1 \partial_\zeta^f(\mathbf{R}_1 \cdot) \\ -\mathbf{R}_1^\top \partial_\zeta^f(G_1^\top \cdot) & r_s \partial_\zeta^f(g_s \cdot) + g_s \partial_\zeta^f(r_s \cdot) \end{bmatrix} \begin{bmatrix} \delta_x \mathcal{H} \\ \delta_s \mathcal{H} \end{bmatrix},$$

where $\partial_\zeta^f(g) := f(x) \partial_\zeta(f(x)g)$.

$\mathbf{R}_0(\mathbf{x}), \mathbf{R}_1(\mathbf{x}) \in \mathbb{R}^{m \times 1}$, $r_s(\mathbf{x}) \in \mathbb{R}$ are the modulating functions defined as follows

$$R_{0,i} = \gamma_{0,i}(\mathbf{x}, \delta_x \mathcal{H}, \delta_s \mathcal{H}) \{S|G_0(:, i)|\mathcal{H}\},$$

$$R_{1,i} = \gamma_{1,i}(\mathbf{x}, \delta_x \mathcal{H}, \delta_s \mathcal{H}) \{S|G_1(:, i)|\partial_\zeta^f \mathcal{H}\},$$

$$r_s = \gamma_s(\mathbf{x}, \delta_x \mathcal{H}, \delta_s \mathcal{H}) \{S|g_s(x)|\mathcal{H}\}.$$

The pseudo Poisson brackets are defined as

$$\{\mathcal{E}|\mathcal{G}|\mathcal{F}\} = \begin{bmatrix} \delta_{\mathbf{x}} \mathcal{E} \\ \delta_s \mathcal{E} \end{bmatrix}^\top \begin{bmatrix} 0 & \mathcal{G} \\ -\mathcal{G}^* & 0 \end{bmatrix} \begin{bmatrix} \delta_{\mathbf{x}} \mathcal{F} \\ \delta_s \mathcal{F} \end{bmatrix},$$

$$\{\mathcal{E}|g_s(x)|\mathcal{F}\} = \delta_s \mathcal{E} \partial_\zeta(g_s(x) \delta_s \mathcal{F})$$

for some smooth function $\mathcal{E}$ and $\mathcal{F}$, operator $\mathcal{G}$ and its formal adjoint $\mathcal{G}^*$, and a real continuous scalar function $g_s(x)$.

Boundary inputs and outputs

Assumption

The operator $\begin{bmatrix} \delta_x \mathcal{H} \\ \delta_s \mathcal{H} \end{bmatrix}^\top \begin{bmatrix} P_1 \partial_\zeta^0(\cdot) \\ -G_1^\top C(\mathbf{x}) \end{bmatrix} \cdot \delta_x \mathcal{H}$ is an exact differential form.

A boundary controlled IPHS is an infinite-dimensional IPHS with the boundary port variables

$$u(t) = W_B \begin{bmatrix} e(t, b) \\ e(t, a) \end{bmatrix}, y(t) = W_C \begin{bmatrix} e(t, b) \\ e(t, a) \end{bmatrix},$$

where

$$e(t, \zeta) = \begin{bmatrix} f_0(x) \frac{\delta \mathcal{H}}{\delta x}(t, \zeta) \\ f(x) \mathbf{R}(\mathbf{x}, \frac{\delta \mathcal{H}}{\delta \mathbf{x}}) \frac{\delta \mathcal{H}}{\delta s}(t, \zeta) \end{bmatrix},$$

$$\mathbf{R}\left(\mathbf{x}, \frac{\delta \mathcal{H}}{\delta \mathbf{x}}\right) := \left[\mathbf{1} \mathbf{R}_1\left(\mathbf{x}, \frac{\delta \mathcal{H}}{\delta \mathbf{x}}\right) \mathbf{r}_s\left(\mathbf{x}, \frac{\delta \mathcal{H}}{\delta \mathbf{x}}\right) \right]^\top,$$

$$W_B = \left[\frac{1}{\sqrt{2}} (\Xi_2 + \Xi_1 P_{ep}) M_p \frac{1}{\sqrt{2}} (\Xi_2 - \Xi_1 P_{ep}) M_p \right],$$

$$W_C = \left[\frac{1}{\sqrt{2}} (\Xi_1 + \Xi_2 P_{ep}) M_p \frac{1}{\sqrt{2}} (\Xi_1 - \Xi_2 P_{ep}) M_p \right],$$

where $M_p = (M^\top M)^{-1} M^\top$, $P_{ep} = M^\top P_e M$, $M \in \mathbb{R}^{(n+m+2) \times k}$ is spanning the columns of $P_e \in \mathbb{R}^{n+m+2}$ of rank k , defined by

$$P_e = \begin{bmatrix} P_1 & 0 & G_1 & 0 \\ 0 & 0 & 0 & g_s \\ G_1^\top & 0 & 0 & 0 \\ 0 & g_s & 0 & 0 \end{bmatrix},$$

$\Xi_1, \Xi_2 \in \mathbb{R}^{k \times k}$ such that $\Xi_2^\top \Xi_1 + \Xi_1^\top \Xi_2 = 0$, $\Xi_2^\top \Xi_2 + \Xi_1^\top \Xi_1 = I$.

Boundary inputs and outputs for convective fluids in 1D

Eulerian coordinate: The boundary inputs and outputs are

$$\mathbf{u} = \begin{bmatrix} -(\frac{1}{2}v^2 + h)(b) - \frac{1}{\rho}\tau(b) \\ -f_s(b) \\ (\frac{1}{2}v^2 + h)(a) + \frac{1}{\rho}\tau(a) \\ f_s(a) \end{bmatrix}, \mathbf{y} = \begin{bmatrix} \rho v(b) \\ T(b) \\ \rho v(a) \\ T(a) \end{bmatrix},$$

where $f_s := -\frac{k}{T}\partial_\zeta T = \frac{q}{T}$, such that

$$\dot{\mathcal{H}} = \mathbf{y}^\top \mathbf{u}, \dot{S} = \int_a^b \sigma_s d\eta + \mathbf{y}_s^\top \mathbf{u}_s,$$

where $\mathbf{y}_s^\top \mathbf{u}_s := -(s\rho v + \frac{q}{T})|_a^b$,

$$\sigma_s := \frac{1}{T^2} k(\partial_\zeta T)^2 + \frac{\mu}{T} (\partial_\zeta v)^2 \geq 0.$$

Note: In the general form, $C(\mathbf{x}) = \partial_\zeta s$, $f_0 = 1$, $f = \frac{1}{\rho}$.

Lagrangian coordinate: The boundary inputs and outputs are

$$\mathbf{u} = \begin{bmatrix} -p(b) - \tau(b) \\ -f_s(b) \\ p(a) + \tau(a) \\ f_s(a) \end{bmatrix}, \mathbf{y} = \begin{bmatrix} v(b) \\ T(b) \\ v(a) \\ T(a) \end{bmatrix},$$

where $f_s := -\frac{k}{T}\partial_\zeta T = \frac{q}{T}$, such that

$$\dot{\mathcal{H}} = \mathbf{y}^\top \mathbf{u}, \dot{S} = \int_a^b \sigma_s d\eta + \mathbf{y}_s^\top \mathbf{u}_s,$$

where $\mathbf{y}_s^\top \mathbf{u}_s := -(\frac{q}{T})|_a^b$,

$$\sigma_s := \frac{1}{T^2} k(\partial_\eta T)^2 + \frac{\mu}{T} (\partial_\eta v)^2 \geq 0.$$

Note: In the general form, $C(\mathbf{x}) = 0$, $f_0 = f = \frac{1}{\rho_0}$.

Conclusion and outlook

- We introduced the irreversible port-Hamiltonian formalism for convective fluids both in fixed and moving frame.
- We extended the previous form of BC-IPHS to include the convective transport via appropriate modification of the differential operators.
- We also introduced the parametrization of the boundary port variables for consistent consideration of thermodynamic laws.

Contribution:

- A. Ouardi, A. Sarkar, H. Ramirez, Y. Le Gorrec: *Irreversible Port-Hamiltonian Formulations for 1-Dimensional fluid systems*, submitted to **IFAC World Congress**, 2026.

Outlook

- Identification of a regular structure of $C(\mathbf{x})$ to bypass the assumption.
- Considerations of port maps essential for interconnection of infinite-dimensional and boundary controlled IPHS, necessary for potential applications, e.g., heat exchangers, heat pumps, etc.
- Comparison of the BC-IPHS with other formalisms such as GENERIC for open systems.

THANK YOU!